

KUTAISI INTL, Georgia

WMO: 371775

Lat: 42.177N

Lon: 42.483E

Elev: 223

StdP: 14.58

Time Zone: 4.00 (E04)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.1	30.3	19.1	14.5	41.3	21.3	16.1	42.1	38.8	48.8	34.7	49.2	7.6	90	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.7	94.9	73.7	91.4	73.6	88.1	72.5	78.3	88.0	76.8	85.7	75.5	83.5	17.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	134.2	83.2	73.8	126.8	81.5	73.2	124.3	80.8	41.8	87.9	40.4	86.1	39.1	84.4	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
39.0	32.2	27.1	DB	22.6	101.2	4.8	3.4	19.1	103.6	16.3	105.6	13.6	107.5	10.1	110.0	
			WB	22.2	81.1	5.1	2.5	18.6	82.9	15.6	84.4	12.7	85.9	9.0	87.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.8	44.3	45.8	50.6	59.0	66.4	73.1	76.5	77.9	71.7	62.9	53.5	46.6
	DBStd	13.53	6.12	7.48	7.92	7.10	6.18	4.61	4.56	4.54	6.48	6.29	7.11	7.49
	HDD50	648	194	155	92	6	0	0	0	0	1	46	154	
	HDD65	2948	643	538	449	204	59	2	0	21	116	346	570	
	CDD50	4579	16	38	110	278	508	692	821	864	651	401	151	49
	CDD65	1403	0	0	2	26	102	244	356	399	222	51	1	0
	CDH74	11920	0	1	17	336	933	1941	2967	3612	1745	344	23	1
	CDH80	4399	0	0	0	69	297	663	1145	1529	636	60	0	0
Wind	WSAvg	9.6	10.6	12.7	11.8	11.0	8.3	7.9	7.2	7.6	8.2	9.2	10.3	10.5
Precipitation	PrecAvg	56.7	5.4	5.0	3.9	3.3	2.5	4.4	3.3	4.4	5.2	6.0	6.0	7.4
	PrecMax	75.2	12.1	11.8	7.9	9.8	4.3	10.6	9.7	13.8	11.2	12.2	17.9	15.9
	PrecMin	42.0	1.1	0.5	1.5	0.8	0.6	1.8	0.6	0.4	1.3	0.6	0.6	1.7
	PrecStd	8.5	2.9	2.8	1.7	1.7	1.0	1.9	2.2	2.4	2.4	3.2	3.7	3.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.8	71.4	76.7	86.1	91.4	94.9	100.4	100.5	96.9	85.8	77.0	70.1
		MCWB	47.6	52.9	55.6	65.1	69.2	72.2	74.7	70.4	69.1	66.0	58.1	55.4
	2%	DB	59.4	66.2	71.7	81.0	87.5	89.9	94.6	94.8	91.0	80.9	72.0	64.6
		MCWB	47.0	49.9	54.3	61.4	67.5	72.7	74.2	75.1	69.5	64.9	56.6	50.7
	5%	DB	57.2	62.2	68.0	77.4	82.7	86.3	89.6	91.3	86.2	77.4	68.3	61.0
		MCWB	45.5	49.4	53.1	60.3	66.5	71.5	75.4	75.4	69.7	62.9	55.0	49.3
	10%	DB	54.0	57.3	64.1	73.5	79.0	84.0	86.0	87.9	82.8	74.8	64.7	57.4
		MCWB	44.5	47.3	51.4	58.1	65.4	70.7	74.8	74.3	69.3	63.2	54.0	48.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.1	55.5	57.7	66.3	72.3	77.5	80.4	80.9	75.5	69.2	61.6	56.7
		MCDB	57.7	67.3	70.5	82.8	84.6	86.6	88.7	91.1	86.3	79.0	69.9	65.0
	2%	WB	49.1	52.1	55.5	63.5	69.8	75.5	78.5	78.5	73.4	67.3	59.2	53.6
		MCDB	54.3	61.4	68.3	78.4	82.0	83.9	87.8	88.6	82.8	76.3	66.0	61.4
	5%	WB	47.4	50.1	54.0	61.3	67.9	74.0	76.8	77.1	72.2	65.5	57.4	51.1
		MCDB	53.1	57.9	65.5	74.8	79.9	82.3	85.5	86.3	81.4	73.9	66.3	57.9
	10%	WB	45.8	48.3	52.1	59.2	66.2	72.3	75.3	75.8	70.8	63.9	55.5	49.2
		MCDB	51.4	55.3	62.2	71.4	77.1	80.7	83.1	84.0	79.6	72.0	62.8	55.6
Mean Daily Temperature Range	5% DB	MDBR	13.8	14.9	16.4	21.1	19.6	18.9	17.0	18.7	19.1	18.3	16.8	14.8
		MCDBR	18.6	22.8	23.9	29.2	27.1	23.9	23.1	23.8	24.9	23.8	24.4	19.9
		MCWBR	10.4	11.4	11.4	14.3	13.4	10.9	10.3	10.1	10.8	11.5	12.8	11.6
	5% WB	MCDBR	15.0	19.2	20.5	26.5	24.3	21.1	19.0	20.1	21.5	20.5	21.0	17.6
		MCWBR	9.4	11.1	10.9	14.0	12.8	10.7	10.0	9.6	10.6	11.5	11.9	11.1
Clear-Sky Solar Irradiance	taub	0.341	0.376	0.433	0.486	0.452	0.438	0.467	0.471	0.413	0.403	0.357	0.334	
	taud	2.273	2.153	2.058	1.995	2.180	2.288	2.230	2.214	2.359	2.322	2.398	2.343	
	Ebn at Noon	248	256	254	250	263	266	257	252	259	245	242	240	
	Edh at Noon	29	38	47	54	46	42	43	43	35	32	26	25	
All-Sky Solar Radiation	RadAvg	525	793	1059	1429	1685	1864	1839	1725	1394	942	633	462	
	RadStd	72	92	92	164	168	144	174	130	133	104	83	47	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page